

PAPER 3 — HANDOFF v11.0

Written 24 August 2026, end of session. Supersedes v10.0 where they disagree. Keep v10.0 alongside: it holds the Block 2/3 narrative, the training-pair format and the diagonal-blob finding, none of which is repeated here in full.

0. HOW TO USE THIS FILE

This session was almost entirely about **data validity**: five separate faults were found and fixed, each of which was silently producing or certifying bad grasps. Section 2 is the important one. Section 4 says what it means for data already on disk.

Verify file freshness before editing. Uploads to a new chat are not guaranteed to match disk. Grep for a known signature first:

File	Signature to grep for
main_gui.py	THE TRANSPOSITION, FIXED HERE AND ONLY HERE
main_gui.py	grid_gripper_cloud_gap_mm
sim/collect_from_config.py	contact_at
examples/probe_finger_throat.py	tool_basis_from_pads
Data/finger_throat.json	cloud_xyz_mm

1. BIG PICTURE (unchanged)

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Block	What	Status
1	Grid data collection in Isaac Sim	Complete; five validity faults fixed today
2	Stitching 7x4 maps into an extended contact map	Complete, validated
3	U-Net contact completion (initial -> extended)	Prototyped; awaiting a larger dataset
4	A/B against Paper 1's shape rules	Not started

Hardware/sim: UR5e + Robotiq 2F-85, two TSF-85 pads (7x4 = 28 taxels). Isaac Sim 5.1, TSF-85 extension by Berith Atemoztli De la Cruz Sanchez, cuRobo for planning. Ubuntu 22.04, RTX 2060.

2. WHAT CHANGED THIS SESSION

Five faults, found in this order. Each was found because the previous fix exposed it.

2.1 The nx / ny transposition (every automatic grid was wrong)

design_grid derives one count from the ACROSS contact band and one from the ALONG-rod window. **grid_2d**'s first count steps along world Z (UP the rod) and its second across world Y. **do_design_grid** handed them over in the order they were derived — transposed.

Proof, from your own config. run_20260821_165542 (D42, 25 deg): observed vertical excursion 29.358 mm equals $24 \cdot \cos 25 + 18 \cdot \sin 25$ to 0.001 mm, which only holds if the count derived from the across band was driving the vertical steps.

Cost, both real. The vertical sweep overran its own window (feeding fault 2.2). And on the upright runs the across sweep reached ± 18 mm against a designed ± 15.53 , leaving **0.53 mm** of pad inside the contact band at the outermost column — near-empty maps. On D26 that is 10 of 35 points.

Fixed at the hand-over point only. design_grid now returns n_across / n_along by name; do_design_grid maps n_along to the X box and n_across to the Y box. grid_2d is deliberately unchanged: its convention is what the collector, the stitcher, the serpentine order and every run on disk already speak.

2.2 Per-point gripper-vs-rod check in the grid designer

The four limits in design_grid are scalars applied to the base height. Two things slipped through once the pad was rolled: the palm limit measured the lever arm as if the tool were vertical (over-stating clearance by 6.5 mm at 25 deg, 20 mm at 45 deg), and the across steps carry a vertical component of $n_across \cdot \text{step} \cdot \sin(\text{roll})$ that the along window never saw. On the D42 @ 25 deg run, 7 of 63 points put the rod **inside** the gripper body, worst -3.15 mm.

Every point is now verified against the real gripper at its own rolled tool pose. If a grid fails, the base height is re-centred first, a step is trimmed only if that is not enough, and it **refuses** rather than emit a colliding grid. At 0 deg this reproduces the old scalar palm rule to 0.01 mm, so upright designs are unchanged.

2.3 The D60 jam — the rod's top end wedges in the fingers

A single D60 grasp at pad_dz +29.64 returned **peak 4** counts on s1 and 1 on s2 — no contact — while the same pose at +50.0 gave peak sum 13617, and the stored calibration (peak sum 14314) was taken at +39.15. Roll cannot explain it: the roll axis IS the closing axis, so a rolled pad squeezes exactly as hard as an upright one.

What separates the three is how far the rod's TOP END pokes up inside the gripper, measured past the flange along the tool axis:

rod top past flange	outcome	pose
110.6 mm	DEAD (peak 4)	pad_dz +29.64
120.1 mm	firm (peak sum 14314)	the pose ALL NINE calibrations were taken at
131.0 mm	firm (peak sum 13617)	pad_dz +50.0

That every calibration in the file sits at 120.0–120.1 mm is not a coincidence — they were all taken at the same fixed EE height, and pad_dz drifting +32.8 to +39.15 across diameters is just TOOL_OFFSET_Z shrinking. It also means the calibration file has never sampled the jam region.

Two hypotheses were tested and killed first, recorded so they are not re-tried: (a) knuckle blocking, rejected because the cuRobo sphere set flags 2 knuckles at 0 deg and 4 at 25 deg, i.e. backwards; (b) marginal calibration, rejected because close_rad sits on a straight line through all nine diameters to 0.0003 rad and every peak sum is 8600–15400.

2.4 probe_finger_throat.py — measuring the gripper

New one-shot probe. For each calibrated close_rad it drives the fingers to that angle in free air (object scaled to 1 mm so it cannot stall them) and exports the gripper as a point cloud in the tool frame, plus a diagnostic radial profile. Headless, all nine diameters, ~2 min.

The frame is measured, not assumed. The two pads straddle the tool axis, so flange-to-pad-midpoint is the approach axis and the line joining them is the closing direction. Validation: pad midpoint depth matches TOOL_OFFSET_Z to **0.4 mm on all nine diameters** (151.12 vs 150.99; 156.87 vs 156.57; 157.56 vs 157.22), and pad separation tracks diameter with slope 1.

Shallowest depth a rod of each diameter may occupy:

D (mm)	10	13	20	26	32	42	45	52	60
min depth (mm)	13	14	102	102	102	104	104	110	112

It predicted the failure it was written to explain. D60 comes out at 112.0 mm, against a dead grasp at 110.6 and a firm one at 120.1 — a 1.4 mm call, made without access to those numbers.

2.5 How the check actually works (and three wrong versions)

Worth recording, because each wrong version looked reasonable:

1. **One radius per depth.** Wrong: the throat is not circular. It is narrow across the fingers and wide open the other way, so a rod stepped along the OPEN direction scored as if driven into a finger. It rejected the D60 @ 25 deg grid that had actually grasped (peak 395) and shrank every upright grid to 3 points.
2. **Unsigned distance.** Wrong: clamping at zero reports a point INSIDE the rod as 0.00 mm. The dead pose has exactly one gripper point inside the object, and it passed.
3. **Fingers left in.** Wrong: they sit 1–2 mm off the rod at every valid grasp, so the nearest gripper point is always a holding surface and the jam signal is buried. D26 at its six-run base: +3.12 mm with fingers in, +13.88 with them out.

What works: exclude any gripper point at or beyond the WORKING RADIUS (half the measured pad separation) — that is what makes a surface a holding surface — then take the SIGNED distance from each remaining point to the rod solved analytically. A depth cut was tried and does not work: the finger's inner face runs at the working radius along its whole length.

The margin is bracketed by observation, not chosen:

case	gap	meaning
D60 pad_dz +29.64	-0.06 mm	peak 4 — DEAD
D26 pad_dz +16.31	+1.88 mm	lowest point of the six upright runs
D60 pad_dz +39.15	+3.13 mm	peak sum 14314 — the calibration pose

The boundary lies between -0.06 and +1.88, so anything at or above 1.88 would reject grasps that demonstrably worked. **THROAT_MARGIN_MM = 1.0.** Widen it only against new evidence.

2.6 The gripper is finally in cuRobo's collision model

ur5e_gripper.yml (body, coupler, knuckles; fingers and pads deliberately absent) existed and was validated on 7 August, then never switched on. Every run until today logged **gripper_modelled: false** while the collector printed its own warning each time. There is now a checkbox in the Collection tab, **on by default**, feeding both the reachability and the run command.

Effect: the model goes from 28 spheres reaching 51 mm past the flange to **45 spheres reaching 133 mm**.

2.7 The ascent contact zone (exposed by 2.6)

plan_stitched_z releases the object obstacle within 30 mm **of the goal** — correct for a descent, where the goal is the contact. The ascent reuses the same function with the contact at the START, so it released at the top and held the obstacle on at the bottom, where the gripper is still wrapped round the rod. cuRobo rejected it before planning: *Start or End state in collision*.

Fixed with a **contact_at** parameter defaulting to "end", so the three descent callers are bit-identical; only the ascent passes "start". 30 mm is enough to clear: the gripper's spheres reach down to Z 1113.4 at the grasp and the inflated obstacle tops out at 1127.2, so 13.8 mm suffices. Confirmed on a full D60 grid.

2.8 GUI quality of life

- **Stitch tab scrolls.** The blob-axis block and status line were below the window edge with no way to reach them.
- **Heatmaps save only.** The per-grasp matplotlib window loop is gone; a 63-point sweep used to bury the screen in 63 windows.
- **Gripper-collision checkbox** (see 2.6).

3. THE VALIDATION RUN

run_20260824_151540_obj0_pad0 — D60 x 140, 0 deg roll, auto-designed grid, gripper collision ON. This is the run that proves the whole chain, on the exact case that produced peak 4.

Quantity	Result
Grid	21 points (3 across x 7 along), pad_dz +43.16, step 6 mm
Ledger	robot_yaml ur5e_gripper.yml, gripper_modelled true, 45 spheres / 133 mm
Reachability	21/21 OK; 21/21 executed; 0 false positive, 0 false negative
Contact	s1 peaks 652–1407, s2 547–1390 — not one dead map
Blob axis	mean error -0.20 deg (s1), +0.14 deg (s2); spread ~2–3 deg
Band fit	22.0 mm -> equivalent indentation 2.09 mm
Overlap sigma	111.4 (s1) / 101.8 (s2)
Ascent	pt20 completed after the 2.7 fix

Two earlier claims this retracts. (a) The "grip is far lighter than assumed" conclusion — 0.30 mm equivalent indentation — came from the D60 @ 25 deg run and was an artifact of the jam; on a proper grasp the fit gives 2.09 mm against the 2.4 mm design_grid assumes, so the assumption is roughly right. (b) The s1/s2 2.3x asymmetry, seen twice and called systematic, is gone at 1407 vs 1390 — it was a symptom of the marginal grasp, not a sensor fault.

Caveat on the band fit: it railed at 22.0 mm, the top of the search range, so 2.09 mm is a lower bound. The report also prints its own large-residual warning.

4. WHAT THIS MEANS FOR DATA ALREADY ON DISK

Nothing collected before today is corrupt, but two things must be stated whenever it is used:

- The six upright runs were collected with a transposed grid.** The sweep is still a valid sweep, but the across extent ran to ± 18 mm against a designed ± 15.53 , so the outermost across column grasped an 0.53 mm sliver of the contact band. Those points are near-empty maps, honestly recorded in each gui_config_used.json.
- Any run whose rod top sat shallower than min_depth is suspect.** Check it as TOOL_OFFSET_Z - (L/2 - pad_dz) against the table in 2.4. The D60 sweeps of 21–23 August are the known bad ones.

Design heights change slightly from here: D26 upright moves from pad_dz +28.31 to +33.43, because the old base put the lowest points of the sweep near the jam boundary. Grids will therefore not reproduce the six earlier runs exactly. This is a decision already taken — the new number is the correct one.

5. VERIFIED CONSTANTS

Constant	Value	Provenance
PALM_DROP_MM	86.69	measured: 10.79 flange->palm + 75.9 housing drop
PALM_RADIUS_MM	54.0	NOT measured — cuRobo sphere envelope; fallback only, unused when finger_throat.json exists
THROAT_MARGIN_MM	1.0	bracketed by observation (see 2.5)
CONTACT_ZONE_M	0.030	unchanged; now applied from the contact end
indent (design_grid)	2.4 mm	supported by the 2.09 mm fit on the validation run
HOLD_FRAC / baseline	0.9 / True	unchanged
PAD_W, PAD_H	22.0, 37.0 mm	unchanged
PITCH_Y, PITCH_Z	5.5, 5.2857 mm	unchanged
row 0	pad's physical BOTTOM	proven 2026-07-22

Tool frame, confirmed by Isaac itself this session. The run log prints tool axis in world = [0.0, 0.4226, -0.9063] at 25 deg, which is exactly (0, sin25, -cos25), and pivot correction (+0.0, -63.8, -14.1) mm, which is (0, -151·sin25, -151·(1-cos25)). The matrix the grid check is built on is therefore verified against the simulator, not just derived.

6. KNOWN-BROKEN / NOT YET DONE

Item	Severity	Note
Diagonal blob artifact	High	The CNN renders 0 and 90 deg faithfully but collapses diagonals. Berith has confirmed it is real and is working on it (new, this session) . Rolled/tilted data remains unusable for training until it lands; 0 and 90 deg are the two orientations that are safe today.
Closing is open-loop; nothing verifies contact	High	Every grasp closes to a fixed close_rad taken from the calibration file — a hand-picked angle per diameter, with a D26 fallback for any size not in the file. Nothing checks that contact happened, so a dead grasp is only visible afterwards in the peaks. Blocks unattended batch collection, and is the root of the guesswork the contact-aware close is meant to remove.
The collector cannot read tactile in-process	Medium	It talks to the TSF-85 extension through carb settings and only WRITES them (record_active, sensor_root, output_dir, base_name). Every tactile number comes from the CSV afterwards. This is what stands between here and a close-until-contact loop — see section 9 for the three ways round it.
Hand-typed grids skip the throat check	Medium	Design Grid refuses a 1-point sweep, so single-point tests are typed by hand and bypass every limit. Keep pad_dz above the table in 2.4.
Obstacle off for the first 30 mm of ascent	Low	Unavoidable while the fingers are round the object; the same trade the descent already makes at the other end.
validation.py not updated for rolled pads	Low	_sample_canvas still assumes an axis-aligned footprint. Correct on flat runs. Only matters when rolled data returns.
GSR saturated	Low	Zero dynamic range on trivially successful grasps; do not quote it as validation.
PALM_RADIUS_MM unmeasured	Low	Now fallback-only. Caliper it if the throat file is ever unavailable.

7. FILE INVENTORY

File	Change this session
main_gui.py	transposition fix; per-point gripper check using the measured throat; gripper-collision checkbox; Stitch tab scrollbar; heatmaps save-only
sim/collect_from_config.py	plan_stitched_z contact_at parameter (ascent)
examples/probe_finger_throat.py	NEW — measures the gripper throat
Data/finger_throat.json	NEW — 9 diameters, ~4900 points each, ~3 MB
Data/pad_offset_calibration.json	unchanged; all nine entries verified consistent
viz/stitching.py, blob_axis.py	unchanged

Key runs. run_20260821_165542 (D42 25 deg — proved the transposition); run_20260823_145830 (D60 25 deg, single — peak 395); run_20260823_150843 (D60 0 deg, single — peak 4, the jam); run_20260824_151540 (D60 0 deg, 21 points — the validation run).

8. CLOSING THE LOOP — WHAT IS AND IS NOT AVAILABLE

Recorded because it was checked this session and it decides how the contact-aware close gets built.

The collector runs INSIDE Isaac Sim, in the same process as the sensor. A closed-loop close therefore needs no streaming, no ROS2 bridge and no server — it can read the signal directly. Streaming is only needed for a SEPARATE live window. Those two are independent problems and should not be solved together.

Three candidate signals for "close until contact", cheapest first:

Signal	Available in-process?	Notes
Geometric indentation D/2 - pad centre - rod axis	Yes, today	The calibrator already reads these pad primis at closed grip. Exact in sim. Assumes the object's pose and diameter are known, so it does not transfer to the real robot.

Signal	Available in-process?	Notes
Pad deformation max sim-rest on simulation_mesh	Yes, today	Physical, CNN-free, and demonstrably live: the run log shows it go 0.000000 -> 0.000127 -> 1.154909 -> 1.157376 during a close. Independent of both the calibration file and Berith's CNN, and the same idea works on a real sensor. Recommended.
Tactile counts (CNN output)	No	Would need the extension to expose the current prediction, or a CSV tail mid-close. Also inherits the CNN's own open questions.

For a live view (a second window showing both tactile maps and the finger pose relative to the object) the options are: tail the CSVs the extension already writes — no new infrastructure; Berith's existing Qt server — exists, but its 2.5 Hz vs 60 Hz rate is one of his open questions; or Isaac's ROS2 bridge — heaviest, but gives one viewer that serves both sim and the real UR5e, which ran ROS2 Humble in Paper 2.

The case for it: today's jam was invisible precisely because runs are headless, and a live map would have shown peak 4 in seconds rather than after a full sweep and a stitch. The case against: unattended batch collection means nobody is watching anyway, and a per-run log catches the same thing.

9. WORKING PREFERENCES (carried forward)

- One verifiable step at a time. Measure, don't guess.
- Distances in mm. Complete files, never diffs. Commands on one line.
- **Kourosh's visual read of the simulation overrides diagnostics when they conflict.** This has been correct every time.
- Never present schematic pseudocode as a quotable line from a real file.
- Verify file freshness before editing (section 0).

*A note this session earned: **three of my implementations of the throat check were wrong**, and each was caught by comparing against grasps that had already worked rather than by reasoning. The pattern worth keeping is that every new limit must be validated against existing data before it is trusted to reject anything.*